Conduction inside the circuit

1 Problem Setup

The conductor is one closed loop, divided into wire, one battery, and possibly several resistor regions. Let ρ be the resistivity in each region and $\gamma = 1/\rho$ its conductivity. The electrostatic field is $\boldsymbol{E} = -\nabla V$. The battery also pushes charge by a mechanism represented by a driving field $\boldsymbol{F}$. The current density is

$$\boldsymbol{J} = \gamma(\boldsymbol{E} + \boldsymbol{F}) = \gamma(-\nabla V + \boldsymbol{F}). \tag{1}$$

If the battery has EMF $\mathcal{E}$, length ℓ_b, and unit driving direction $\boldsymbol{d}_b$, the model sets

$$\boldsymbol{F} = \begin{cases} (\mathcal{E}/\ell_b)\boldsymbol{d}_b, & \text{inside the battery,} \\ 0, & \text{in the wire and resistors.} \end{cases} \tag{2}$$

The EMF is the total work supplied per unit charge across the battery; $\boldsymbol{F}$ is that drive per unit length. It is separate from the electric field $\boldsymbol{E}$ made by charges.

In steady state $\nabla \cdot \boldsymbol{J} = 0$: charge does not build up inside the conductor. Current cannot cross either conductor–air surface, so $\boldsymbol{J} \cdot \boldsymbol{n} = 0$ there. At a material junction, the potential and normal current are continuous. No terminal voltage is prescribed, because the conductor forms a closed loop.

2 Turning conservation into equations

The shared derivation in `fem.tex` starts with $\boldsymbol{a} = c(\nabla u - \boldsymbol{f})$ and $\nabla \cdot \boldsymbol{a} = 0$. Here choose $u = V$, $c = \gamma$, and $\boldsymbol{f} = \boldsymbol{F}$, so $\boldsymbol{a} = -\boldsymbol{J}$. The boundary term from integration by parts vanishes because $\boldsymbol{J} \cdot \boldsymbol{n} = 0$ on both conductor surfaces. Hence, for each allowed test function w,

$$\int_{\Omega_c} \gamma \nabla V \cdot \nabla w \, dA = \int_{\Omega_c} \gamma \boldsymbol{F} \cdot \nabla w \, dA. \tag{3}$$

Triangles on both sides of a material junction share a common edge. Thus the finite-element potential is continuous there, and equation (3) makes the normal currents balance.

3 Choosing a potential reference

If v solves the conductor equations, then $v + C$ also solves them: only its gradient appears. This is why the assembled matrix has a redundant degrees of freedom. The solver fixes one conductor vertex at zero and solves the remaining free-vertex equations. It then chooses a more useful reference

by subtracting the mean potential over the conductor. Let $\mathcal{T}_c$ be the set of conductor triangles. Then

$$\overline{V} = \frac{\sum_{h \in \mathcal{T}_c} A_h (V_{h,0} + V_{h,1} + V_{h,2})/3}{\sum_{h \in \mathcal{T}_c} A_h}, \qquad V \leftarrow V - \overline{V}. \tag{4}$$

For a linear function on a triangle, the average of its three vertex values is its area average. The final shift changes no electric field or current.

4 Fields used by the app

After solving for the potential, the field and current in each conductor triangle are

$$\boldsymbol{E}_h = -\sum_{i=0}^{2} V_{h,i} \nabla N_{h,i}, \qquad \boldsymbol{J}_h = \frac{\boldsymbol{E}_h + \boldsymbol{F}_h}{\rho_h}. \tag{5}$$

Both vectors are constant within a triangle. The app samples these vectors to draw electric-field and current arrows. Air current is zero; the electric field in air is calculated in `exterior.tex`.

Code: `calculations/conduction.py`.